

Cambridge International AS & A Level

CANDIDATE
NAME

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FURTHER MATHEMATICS

9231/43

Paper 4 Further Probability & Statistics

May/June 2020

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Blank pages are indicated.

	Standard achieved		
	Above average	Average	Below average
Scheme A	31	35	22
Scheme B	19	50	43

[illegible]

Find a 90% confidence interval for the difference between the population means of X and Y . [5]

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- 3 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{1}{5}x & 0 \leq x < 2, \\ \frac{2}{15}(5-x) & 2 \leq x \leq 5, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function of X . [3]

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- (b) Find the median value of X . [2]

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(c) Find $E(X^2)$.

[2]

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(d) Find $P(1 \leq X \leq 3)$.

[2]

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- (b)** Use the probability generating function of Y to find $E(Y)$ and $\text{Var}(Y)$. [5]

[illegible]

5 Students at two colleges, A and B , are competing in a computer games challenge.

- (a) The time taken for a randomly chosen student from college A to complete the challenge has a normal distribution with mean μ minutes. The times taken, x minutes, are recorded for a random sample of 10 students chosen from college A . The results are summarised as follows.

$$\Sigma x = 828 \quad \Sigma x^2 = 68622$$

A test is carried out on the data at the 5% significance level and the result supports the claim that $\mu > k$.

Find the greatest possible value of k . [4]

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- (b) A random sample of 8 students is chosen from college B . Their times to complete the same challenge give a sample mean of 79.8 minutes and an unbiased variance estimate of 9.966 minutes².

Use a 2-sample test at the 5% significance level to test whether the mean time for students at college B to complete the challenge is the same as the mean time for students at college A to complete the challenge. You should assume that the two distributions are normal and have the same population variance. [7]

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The biologist decides to carry out a test at the 5% significance level to test whether the nutrients have resulted in an increase in growth.

[illegible]

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